

$$d^2 + 1^2 = 25 \Rightarrow d^2 = 24 \Rightarrow d = \sqrt{24}$$

SECTION A [Multiple choice questions] - 18 marks

QUESTION 1

If $\sin x = -\frac{1}{5}$, $\frac{3\pi}{2} < x < 2\pi$, then $\sec x =$

$$\sec x = \frac{1}{\cos x} = \frac{1}{\frac{4}{5}} = \frac{5}{4}$$

(a) $\sqrt{24}$	(b) $-\sqrt{24}$	(c) $\frac{5}{2\sqrt{6}}$	(d) $-\frac{5}{2\sqrt{6}}$	(e) $\frac{5}{4}$	(f) $-\frac{5}{4}$	(g) None of (a) - (f)
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QUESTION 2

Consider the piece-wise defined function $f(x) = \begin{cases} -x+2 & \text{if } x > 0, \\ \ln(-x) & \text{if } x < 0. \end{cases}$

$$(f \circ f)(3) = f(f(3)) = f(-1) = \ln 1 = 0$$

(a) 0	(b) 1	(c) 3	(d) $\frac{1}{e}$	(e) Undefined	(f) None of (a) - (e)
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QUESTION 3

The solution of $f(t) = \frac{\ln(t-5)}{t^2+1} \leq 0$ is:

$$\begin{aligned} t-5 > 0 & \text{ and } \ln(t-5) \leq 0 \\ t > 5 & \text{ and } e^{\ln(t-5)} \leq e^0 \\ t-5 & \leq 1 \\ t & \leq 6 \end{aligned}$$

(a) (0, 6]	(b) [0, 6]	(c) [5, ∞)	(d) (5, ∞)	(e) $(-\infty, 6]$	(f) $(-\infty, 6)$	(g) (5, 6]	(h) None of (a) - (g)
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$$\Rightarrow t \in (5, 6]$$

QUESTION 4

$$\sec \frac{20\pi}{3} = \sec \left(\frac{18\pi + 2\pi}{3} \right) = \frac{1}{\cos(6\pi + \frac{2\pi}{3})} = \frac{1}{\cos(\frac{2\pi}{3})} = \frac{1}{\cos(\pi - \frac{\pi}{3})} = \frac{-1}{\cos \frac{\pi}{3}} = \frac{-1}{\frac{1}{2}} = -2$$

(a) $-\frac{\sqrt{3}}{2}$	(b) -2	(c) -1	(d) 1	(e) $\frac{\sqrt{3}}{2}$	(f) None of (a) - (e)
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QUESTION 5

For which value(s) of x is $y = \arcsin(4x-1)$ defined?

(a) $x \in [-1, 1]$	(b) $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$	(c) $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$	(d) $x \in [0, 0.5]$	(e) $x \in [0, \pi]$	(f) $x \in \mathbb{R}$
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$$\begin{aligned} -1 & \leq 4x-1 \leq 1 \\ 0 & \leq 4x \leq 2 \Rightarrow x \in [0, \frac{1}{2}] \end{aligned}$$

QUESTION 6

Choose the one single equation that is not equivalent to the rest of the equations, or choose (e):

(a) $e^x = 100 \Rightarrow e^x = \ln 100 = \ln 10^2 = 2 \ln 10$

(b) $x = \ln(\ln 100) \Rightarrow e^x = \ln 100 \Rightarrow x = \ln(\ln 100)$

(c) $e^x = 2 \ln 10 \Rightarrow x = \ln(2 \ln 10)$

(d) $x = \ln 2 + \ln(\ln 10) \Rightarrow x = \ln 2 + \ln(\ln 10)$

(e) The equations in (a) - (d) are equivalent to each other.

QUESTION 7

The solution set of $7 - |3 - x| > 1$ is:

(a) $(-9, 3)$	☒ (b) $(-3, 9)$	(c) ϕ	(d) $(-\infty, -9) \cup (3, \infty)$	(e) $(-\infty, -3) \cup (9, \infty)$	(f) $\mathbb{R} \setminus 3$	(g) None of (a) - (f)
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QUESTION 8

Consider the following statements:

(1) $\arcsin(\sin x)$ is defined on $(-\infty, \infty)$; ✓

(2) $\cos\left(\arccos\frac{\pi}{6}\right) = \frac{\pi}{6}$; ✓

(3) $\arcsin(\sin \pi) = \sin \pi$. = 0 ✓

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- ☒ (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

$$\begin{aligned}
 7 - |3 - x| &> 1 \\
 \Rightarrow 7 - 1 &> |3 - x| \\
 \Rightarrow |3 - x| &< 6 \\
 -6 &< 3 - x < 6 \\
 -9 &< -x < 3 \\
 -3 &< x < 9
 \end{aligned}$$

QUESTION 9

Consider the following statements:

(1) $\frac{\ln x + \ln x^2}{\ln x^3} = 1$; ✓

(2) $\ln a + \ln b - \ln c = 0$ if $\frac{ab}{c} = 1$; ✓

(3) $\ln x + \ln y = \ln(x + y)$. ✗

$$\frac{\ln x + \ln x^2}{\ln x^3} = \frac{\ln x^3}{\ln x^3} = 1.$$

$$\begin{aligned}
 \ln a + \ln b - \ln c &= \ln \frac{ab}{c} = 0 \\
 \Rightarrow \frac{ab}{c} &= 1.
 \end{aligned}$$

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- ☒ (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

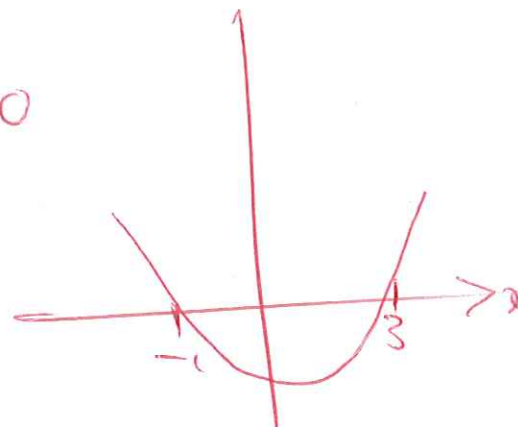
QUESTION 10

The domain of the function $f(x) = \ln\left(\frac{x+1}{x-3}\right) + 1$ is

- (a) $(-\infty, -1) \cup (3, \infty)$;
- (b) $(-\infty, -1) \cap (3, \infty)$;
- (c) $(-1, 3) \cup (3, \infty)$;
- (d) $x \in \mathbb{R} \setminus \{3\}$;
- (e) $(-1, 3)$;
- (f) $(-1, \infty)$;
- (g) $(3, \infty)$;
- (h) None of these.

$$\frac{x+1}{x-3} > 0$$

$$\Rightarrow (x+1)(x-3) > 0$$



QUESTION 11

Consider the function $f(x) = \frac{x-1}{x^2+2x-3}$. Choose the correct statement.

- (a) $x = 1$ is the only vertical asymptote of f ;
- (b) $x = -3$ is the only vertical asymptote of f ;
- (c) f has vertical asymptotes $x = 1$ and $x = -3$;
- (d) f has no vertical asymptotes;
- (e) None of the above.

$$\frac{x-1}{x^2+2x-3} = \frac{(x-1)}{(x-1)(x+3)}$$

$$= \frac{1}{x+3}$$

$$\lim_{x \rightarrow -3^+} f(x) = \lim_{x \rightarrow -3^+} \frac{1}{x+3} = \infty$$

$\Rightarrow x = -3$ vert. asymptote

$$x \rightarrow -3^+$$

$$\Rightarrow x > -3$$

$$\Rightarrow x+3 > 0$$

QUESTION 12

The equation of the graph that results from shifting the graph of

$$y = e^{-x}$$

2 units to the right is:

(a) e^{-x+2}	(b) e^{-x-2}	(c) $e^{-x} - 2$	(d) $2e^{-x+2}$	(e) None of (a) - (d)
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End of Section A

$$y = e^{-(x-2)}$$

$$= e^{-x+2}$$

SECTION B [Fill in questions] - 12 marks

SECTION B: Write only your final, fully simplified answer in the block provided. Only this will be marked.

QUESTION 13 [1 mark]

Simplify the fraction.

$$\frac{(e^{2x})^3(e^x)^2}{e^{-4x}} =$$

$$e^{12x}$$

$$\frac{e^{6x} e^{2x}}{e^{-4x}}$$

$$= e^{6x+2x+4x}$$

$$= e^{12x}$$

QUESTION 14 [1 mark]

Determine the infinite limit.

$$\lim_{x \rightarrow 1^+} \ln(\sqrt{x} - 1) =$$

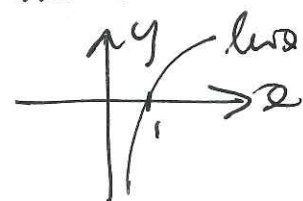
$$-\infty$$

$$x \rightarrow 1^+$$

$$\Rightarrow x > 1$$

$$\Rightarrow \sqrt{x} > 1$$

$$\Rightarrow \sqrt{x} - 1 > 0$$



QUESTION 15 [1 mark]

Solve $|1 + 2x| = 4x$.

$$(1 + 2x)^2 = (4x)^2$$

$$1 + 4x + 4x^2 = 16x^2$$

$$\Rightarrow 12x^2 - 4x - 1 = 0$$

$$\Rightarrow (6x + 1)(2x - 1) = 0$$

$$x = -\frac{1}{6} \text{ or } x = \frac{1}{2}$$

$$\text{But: } x \geq 0$$

QUESTION 16 [1 mark]

Complete the definition of a one-to-one function:

A function f is called a one-to-one function if

See p. 54.

QUESTION 17 [1 mark]

If the average of a, b, c and d is $8t$, write the product $e^a \cdot e^b \cdot e^c \cdot e^d$ in terms of t .

$$e^{32t}$$

$$e^a \cdot e^b \cdot e^c \cdot e^d$$

$$= e^{a+b+c+d}$$

$$\text{Given: } \frac{a+b+c+d}{4} = 8t$$

$$\Rightarrow a+b+c+d = 32t$$

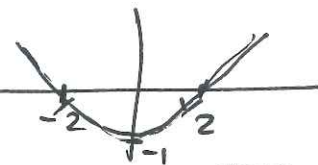
QUESTION 18 [1 mark]

The average rate of change of $f(x) = x^2 + 3$ over the interval $[1, 4]$, is:

$$5$$

$$\frac{f(4) - f(1)}{4 - 1} = \frac{19 - 4}{3} = \frac{15}{3} = 5$$

$$|1-4x^2| = \begin{cases} 1-4x^2 & \text{if } 1-4x^2 \geq 0, 4x^2-1 \leq 0 \\ -(1-4x^2) & \text{if } 1-4x^2 < 0, 4x^2-1 > 0 \end{cases}$$



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QUESTION 19 [2 marks]

Write the function $f(x) = |1 - 4x^2|$ as a piece-wise defined function.

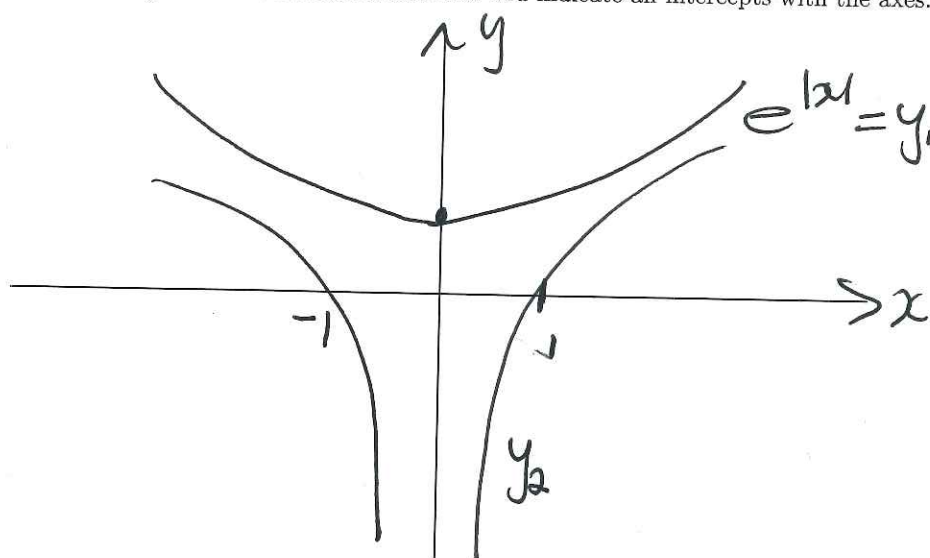
$$f(x) = \begin{cases} 1-4x^2 & \text{if } x \in [-\frac{1}{2}, \frac{1}{2}] \\ -(1-4x^2) & \text{if } x \in (-\infty, -\frac{1}{2}) \cup (\frac{1}{2}, \infty) \end{cases}$$

QUESTION 20 [2 marks]

Sketch the graphs of

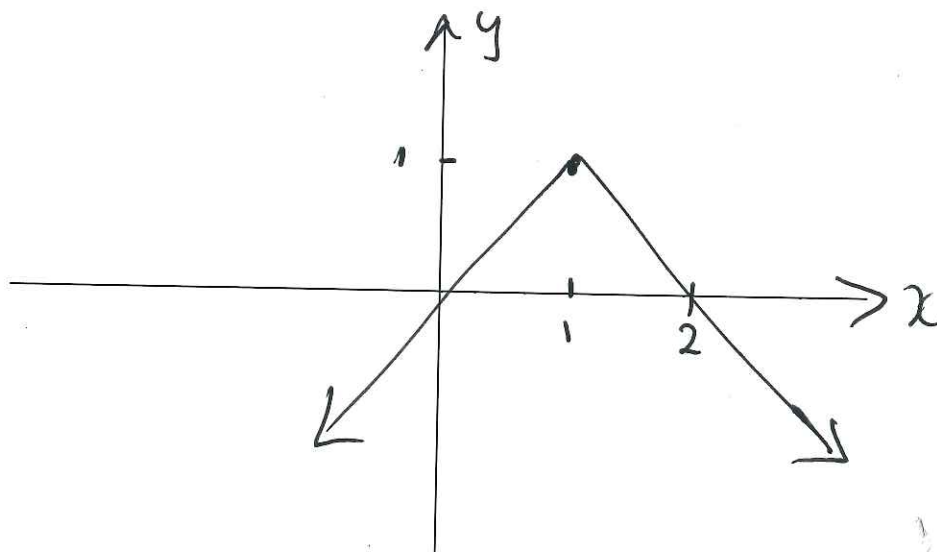
$$y_1 = e^{|x|} \quad \text{and} \quad y_2 = \ln|x|$$

on the same set of axes. Clearly name the functions and axes and indicate all intercepts with the axes.



QUESTION 21 [2 marks]

Sketch the function $f(x) = -|x - 1| + 1$ through transforming the function $g(x) = |x|$. Clearly indicate all intercepts with the axes.



End of Section B

SECTION C [Full written answers] - 20 marks

QUESTION 21 [6 marks]

Solve for x . Write your answer in interval notation.

(a) $\ln x + \ln(x-1) < 0$

$$\ln x(x-1) < 0 \quad \checkmark \quad \text{and:}$$

$$e^{\ln x(x-1)} < e^0 \quad \checkmark$$

$$(4) \quad \boxed{\begin{array}{l} x > 0 \text{ and } x-1 > 0 \\ x > 0 \text{ and } x > 1 \\ \Rightarrow x > 1. \end{array}} \quad \checkmark$$

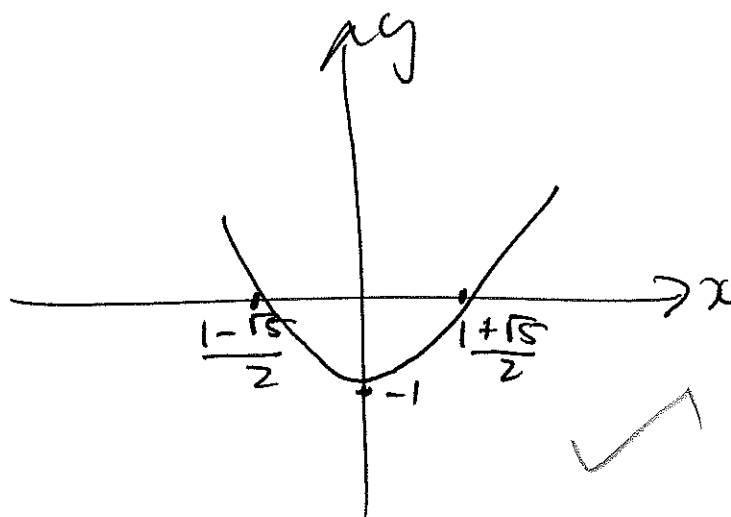
$$\Rightarrow x(x-1) < 1$$

$$\Rightarrow x^2 - x - 1 < 0 \quad \checkmark$$

$$\text{If } x^2 - x - 1 = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{1+4}}{2}$$

$$= \frac{1 \pm \sqrt{5}}{2} \quad \checkmark$$



$$x^2 - x - 1 < 0$$

$$\Rightarrow x \in \left(\frac{1-\sqrt{5}}{2}, \frac{1+\sqrt{5}}{2} \right) \quad \checkmark$$

$$\text{But } x > 1$$

$$\Rightarrow x \in \left(1, \frac{1+\sqrt{5}}{2} \right) \quad \checkmark$$

(b) $1 < e^{3x-2} < 2$. Fully simplify your answer.

(2)

$$\ln 1 < \ln e^{3x-2} < \ln 2$$

$$\Rightarrow 0 < 3x-2 < \ln 2$$

$$\Rightarrow 2 < 3x < \ln 2 + 2$$

$$\Rightarrow \frac{2}{3} < x < \frac{1}{3}(\ln 2 + 2)$$

$$\Rightarrow x \in \left(\frac{2}{3}, \frac{1}{3}(\ln 2 + 2) \right)$$

QUESTION 22 [4 marks]

- (i) Determine if the function
- h
- is even, odd, or neither even nor odd.

(1.5)

$$h(x) = x|x|.$$

$$\begin{aligned} h(-x) &= -x|-x| \\ &= -x|x| \\ &= -h(x) \end{aligned}$$

$\Rightarrow h$ is odd.

- (ii) Suppose
- f
- is an even function and
- g
- is an odd function. Determine if the product
- fg
- is even, odd, or neither even nor odd. Explain your steps.

(2.5)

$$\begin{aligned} f \text{ is even} &\Rightarrow f(-x) = f(x) \\ g \text{ is odd} &\Rightarrow g(-x) = -g(x) \end{aligned}$$

$$\begin{aligned} (fg)(-x) &= f(-x)g(-x) \\ &= f(x)(-g(x)) \\ &= -f(x)g(x) \\ &= -(fg)(x) \end{aligned}$$

$\Rightarrow fg$ is odd.

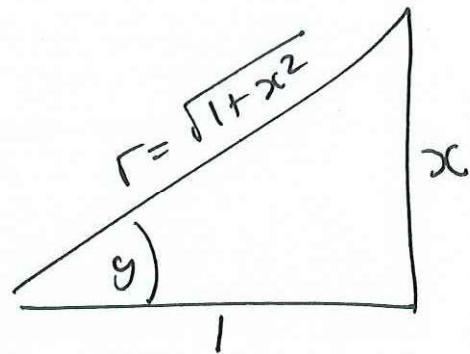
QUESTION 23 [2 marks]

Show how you use the definition of the arctan = $\tan^{-1}$ function to determine the value of

$$\sin(2 \tan^{-1}(x)).$$

Let $y = \tan^{-1}x$
 $\Rightarrow \tan y = x$ and $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$

$$\begin{aligned} & \sin(2 \tan^{-1}x) \\ &= \sin(2y) \\ &= 2 \sin y \cos y \\ &= 2 \frac{x}{\sqrt{1+x^2}} \times \frac{1}{\sqrt{1+x^2}} \\ &= \frac{2x}{1+x^2}. \end{aligned}$$



$$\tan y = x = \frac{x}{1}$$

QUESTION 24 [3 marks]

Consider the function

$$f(x) = \frac{x}{x+2}.$$

Find the inverse function, f^{-1} , together with the domain and range of f and f^{-1} . Use interval notation.

$$y = \frac{x}{x+2}$$

$$\text{Let } x = \frac{y}{y+2}$$

$$\text{then } x(y+2) = y$$

$$\Rightarrow xy - y = -2x$$

$$\Rightarrow y(x-1) = -2x$$

$$y = \frac{-2x}{x-1}$$

$$= \frac{2x}{1-x}$$

$$\Rightarrow f^{-1}(x) = \frac{2x}{1-x}$$

$$D_f = (-\infty, -2) \cup (-2, \infty) = R_{f^{-1}}$$

$$\text{R}_p D_{f^{-1}} = (-\infty, 1) \cup (1, \infty) = R_f.$$

QUESTION 25 [5 marks]

Determine the following limits, if they exist. Do not use L'Hospitals rule.

$$(i) \lim_{x \rightarrow 3} \frac{\sin x}{x} \quad (1)$$

$$= \frac{\sin 3}{3}$$

$$(ii) \lim_{x \rightarrow 3} \frac{\frac{1}{x} - \frac{1}{3}}{x - 3} \quad \left(\frac{0}{0}\right) \quad (2)$$

$$= \lim_{x \rightarrow 3} \frac{\frac{3-x}{3x}}{x-3}$$

$$= \lim_{x \rightarrow 3} \frac{3-x}{3x} \times \frac{1}{x-3}$$

$$= \lim_{x \rightarrow 3} \frac{-(x-3)}{3x} \times \frac{1}{x-3} = \lim_{x \rightarrow 3} \frac{-1}{3x} = -\frac{1}{9}$$

$$(iii) \lim_{x \rightarrow -3} \frac{|x| - 3}{x + 3}$$

$$\left(\frac{0}{0}\right)$$

(2)

$$|x| = -x \text{ if } x < 0$$

$$= \lim_{x \rightarrow -3} \frac{-x - 3}{x + 3}$$

$$= \lim_{x \rightarrow -3} \frac{-(x+3)}{x+3}$$

$$= \lim_{x \rightarrow -3} -1$$

$$= -1$$

End of Section C